

JNTU Hyderabad

MID-TERM EXAMINATION ? 2025

Digital Signal Processing ? Mid-Term Examination, 2025

Subject Code: EC501 Semester: 5 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Discrete Time Signals and Systems)

1. Define Discrete Time Signals and Systems. Explain its significance in Digital Signal Processing with a suitable example. [12 Marks]
2. List the advantages and limitations of Discrete Time Signals and Systems. Explain any two in detail. [12 Marks]
3. Describe the working principle of Discrete Time Signals and Systems with necessary equations. [12 Marks]

SECTION B (Unit 2: DFT and FFT Algorithms)

4. Write short notes on DFT and FFT Algorithms. [12 Marks]
5. Solve a problem related to DFT and FFT Algorithms. Show all the steps clearly. [12 Marks]
6. List the advantages and limitations of DFT and FFT Algorithms. Explain any two in detail. [12 Marks]

SECTION C (Unit 3: FIR Filter Design)

7. Write short notes on FIR Filter Design. [12 Marks]
8. Discuss the engineering applications of FIR Filter Design. [12 Marks]
9. Solve a problem related to FIR Filter Design. Show all the steps clearly. [12 Marks]

SECTION D (Unit 4: IIR Filter Design)

10. Compare and contrast the key concepts involved in IIR Filter Design. [12 Marks]
11. List the advantages and limitations of IIR Filter Design. Explain any two in detail. [12 Marks]
12. Compare and contrast the key concepts involved in IIR Filter Design. [12 Marks]

SECTION E (Unit 5: Digital Signal Processors)

13. Explain Digital Signal Processors with a neat diagram/sketch and a suitable example. [12 Marks]
14. Describe the working principle of Digital Signal Processors with necessary equations. [8 Marks]
15. Derive the expression related to Digital Signal Processors and explain each term.

[12 Marks]

*** End of Question Paper ***